



Department of Computer Science and Engineering

Academic Year 2021– 2022(Odd Semester)

Degree, Semester & Branch: B.E – I Semester Mech

Course Code & Title: GE3151 Problem solving and python programming

Name of the Faculty member (s): Mrs.M.Dhivya AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit III / Iteration statement
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO8
- **Activity Chosen:** Think Pair Share
- **Justification:**
 - Think pair share activity involves the students to think individually and to share the knowledge with classmates.
 - In python, iteration statement like for, while looping discuss syntax and doing simple program.
 - The student's individual solving capability can be tested and the gained ideas can be shared by think-pair-share activity
- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
 - Faculty explained the concept of for loop and while loop.
 - Students were asked to write the answer for the syntax and develop the program for pattern individually within given time as shown in Fig 1.
 - Then the students were asked to discuss the answers with their neighbors.
 - Few students were called in front of the class and asked to share the correct answers after discussion.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO9	PO10	PSO1
CO3	3	1	1	1	1	1	1

(1 – Low 2 – Moderate

3 – High)



PO / PSO mapped:

Innovative practice	Justification for correlation
PO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO9	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PSO1	Students will be able to provide sustainable solution to mechanical engineering problem using the various algorithmic tools.

- Images/ Screenshot of the practice:



Fig 1: Students thinking about the given problem



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- The students felt that this activity is so interesting and helps them to remember the syntax of iteration statement for, while.
- They said it will be very helpful when preparing for exams

- ❖ ***Benefit of the practice:***

- This practice helps the students to remember the concept as they applied what they studied
- It leads to peer to peer learning
- It helps in improving the communication by effectively delivering the things they learnt

- ❖ ***Challenges faced in implementation:***

- All the students were unable to complete the task within allotted time
- Few students found it difficult to solve the expressions individually. After discussion, they felt easy
- Unable to complete the activity within planned time

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/6_IT6801_TPS.pdf
- ❖ <https://www.readingrockets.org/strategies/think-pair-share>
- ❖ <https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-can-improve-your-classroom-discussions1704.html>